Student Sentiment Analysis In Social Media: A Monitoring Tool For Student Satisfaction

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***Abstract*— Sentiment analysis is a method for identifying the emotions or subjective information contained in a piece of text from a variety of sources, such as daily posts and comments on social media by university students. University students often use social media as a forum for freely sharing or making comments about their real ideas and experiences in university life. However, traditional methods of collecting student satisfaction data, such as surveys and feedback forms, have limitations in terms of time and accuracy. With the popularity of social media platforms among students, there is an opportunity to leverage sentiment analysis to gain insights into their opinions and experiences. This study aims to address the limitations of previous research in sentiment analysis and develop a monitoring tool specifically for analysing university student sentiments expressed on social media focusing on University Utara Malaysia case. This study employs the Waterfall Development methodology to develop the proposed model, which consists of five distinct phases. The sentiment analysis model, titled the "UUM Student Satisfaction Analysis Monitoring Tool," was designed based on the requirements, and the evaluation of the effectiveness of the model is done by field testing. USE questionnaire is employed to get feedback from the management authorities of each department at UUM and validate the model's performance. The result of the evaluation was positive feedback from the respondent. The result of F1 score for both sentiments is positive sentiment are displaying 100% accurately detected and negative sentiment are detected 94.83% accurately. It was proven that the UUM Student Satisfaction Analysis Monitoring Tool is valuable for users to analyse the student's opinion on social media to solve real-life problems and improve the management of the university. Although some further improvements are needed, such as enhancing the sentiment analysis dictionary, incorporating clear topic detection, and enabling automatic analysis by file, to make the monitoring tool even more useful and effective.**

Keywords: Sentiment analysis, accuracy, social media

I. INTRODUCTION

Sentiment analysis, which is also known as opinion mining, is a method for identifying the emotions or subjective information contained in a piece of text, such as a phrase, paragraph, or document. It involves the study of the words used to figure out the author's underlying mood or attitude towards a certain subject, services, or product [1]. This analysis of those emotions and viewpoints has made its way into a wide variety of sectors, including consumers details, marketing, publications, apps, websites, and networking. [2]

Daily posts on social media have become a commonplace activity for many individuals, with people from diverse backgrounds and demographics contributing to this digital landscape. Each segment of society finds significance in their online interactions, and one particular group that significantly impacts university life and its operations is the community of university students. Through their active presence on social media platforms, university students bring a distinct and influential perspective that shapes the university's image, communication, and engagement with both the internal and external stakeholders. Their posts on various topics ranging from academics, campus events, and social activities play a pivotal role in shaping the university's online reputation and fostering a vibrant and interconnected campus community.

Universities and colleges basically collect the data of the students' satisfaction by surveys and feedback forms. However, these approaches have some drawbacks as it takes longer time for data collection and feedback is always inaccurate. Due to the increasing popularity of social media platforms like Facebook and Instagram, students now have a new forum for sharing their real ideas and personal experiences about their university life. Researchers have conducted multiple studies on sentiment analysis in order to comprehend the opinions expressed by students on social media platforms. [3] proposed a study using a Lexicon-Based Approach and machine learning to analyse sentiment in Facebook Groups, whereas [4] concentrated on sentiment analysis of students' comments using a similar lexicon-based approach. These studies obtained promising results in determining sentiment on social media and highlighted the benefits of lexicon-based analysis for gaining insights, such as its cost-effectiveness and time-saving nature. Nevertheless, these investigations had some limitations. Inaccurate detection of sentiment-by-sentiment lexical dictionaries, lack of exhaustive text preprocessing techniques, and lack of sources for validating the accuracy of sentiment polarity posed obstacles. Although these works contribute to the field, they do not completely represent the complex details and complications involved in social media sentiment analysis. Therefore, by evaluating these shortcomings in earlier research, the current study aims to remove these limitations and provide a more thorough method to sentiment analysis in the context of social media.

The main aim of this research is to develop a sentiment analysis monitoring tool to identify the accuracy and classification of sentiment of UUM student's opinion that expressed on social media. How can a suitable sentiment analysis model be constructed based on observations of UUM students' sentiments and how can a sentiment analysis monitoring tool be developed to analyse and monitor social media content effectively will be addressed to achieve the objective.

This study introduces a novel sentiment analysis model, titled the "UUM Student Satisfaction Analysis Monitoring Tool," which is designed to monitor student satisfaction by analysing their comments on social media platforms. The proposed model yields percentages representing positive, negative, and neutral words in sentences. The results are then presented both quantitatively and qualitatively, as sentiment scores and sentiment classes respectively. In addition, the model contains a detail section that keeps note of listing of positive and negative terms, as well as prospective sentence topics. This data assists management authorities in effectively prioritising and addressing pressing issues.

This study employs the Waterfall Development (Winston w. Royce,1970) methodology to develop the proposed model, which consists of five distinct phases. To achieve the desired outcomes, the proposed model leverages the capabilities of Stanford Core Natural Language Processing. In order to accurately evaluate the effectiveness of the proposed model and ensure its alignment with the project's objectives and scope, the USE Questionnaire is employed. Feedback from the management authorities of each department at UUM is incorporated to refine and validate the model's performance.

The remainder of this paper is organized as follows: Section 2 will cover some literature review based on sentiment analysis. Section 3 will discuss the process when developing and evaluating the UUM Student Satisfaction Monitoring Tool. Section 4 will analyse the evaluation results and examine them in detail. A discussion of potential future research will follow the conclusion in section 5.

II. Literature Review

The paper [5] discusses the sentiment Analysis of the Uber case on Facebook. This study aims to reveal changes in how Uber is viewed by its users and demonstrate how sentiment analysis can accurately capture customer sentiment and effectively deliver appealing and creative offers. They proposed in the paper that the lexicon-based study categorise the text as positive, negative, or neutral using SVM and Maximum Entropy methods. The outcome of the research shows that the sentiment analysis produced by Classification using SVM approach is 89.01% accurate. In contrast, Maximum Entropy achieved better accuracy of 90.46%. The weakness of the research is lack of comparison with other techniques that potentially provide high accuracy of the polarity of sentiment analysis.

[4] also proposed research on sentiment analysis of students' comments using a lexicon-based approach. The suggested method for determining the sentiment of students' opinions includes utilizing a lexicon-based technique, which separates opinion words into positive and negative terms; using WordNet's dictionary to locate the word's synonyms and antonyms; and using a corpus-based approach to find opinion terms that are relevant to the domain and context. The accomplishment of this study is that it suggested a level of teaching assessment technique that automatically analyses the feedback comments from the students using two different lexicons based on a lexicon-based approach. The limitation of the research is that the dictionary of sentiment detection is relatively small to find the synonym and antonym of the word, which leads to sentiment polarity being inaccurate.

[6] studies the development of a student feedback mining system that can apply and analyse the sentiment of the text from the students' feedback. This study aims to provide instructors with qualitative feedback to improve the students' learning. In this project, stop-word removal is used to cut out the irrelevant words in the text. Next is the clustering process, which partitions the data set into groups based on similar groups and assigns it to each group. Then, the clustered data is classified either positively or negatively by the algorithm. The weakness of this project is that the irrelevant word is not entirely removed from the text, which will lead to the accuracy of the analysis being wrong. In order to prevent this problem, in addition to stopping word removal, other irrelevant words (such as overlapping words) or emoticons will also be removed.

[7] constructed a study titled "Visualizing Student Opinion Through Text Analysis," which involves the use of the Natural Language Toolkit in Python for text preprocessing, the latent Dirichlet allocation (LDA) statistical method to identify sentence aspects, and the AFINN dataset to determine sentiment and calculate sentiment scores. The suggested approaches for achieving the visualization techniques investigated give educators a summary of the elements and emotions they are related to, and it enables educators to quickly compare the courses and teaching time in the same course. This study has a single drawback, which indicates that the text preprocessing was not done very well.

III. Methodology

1. *Project Development Methodology*

This section describes the development procedure for the proposed sentiment analysis monitoring tool utility. Waterfall Development had been employed to develop the "UUM Student Satisfaction Analysis Monitoring Tool". Waterfall Development is the linear approach to complete each phase before moving to the next phase. This methodology is chosen for this research project because it provides a structured and systematic framework for model development. Waterfall approach ensures that each phase is thoroughly completed and documented, minimizing the risk of overlooking important aspects of the development process.

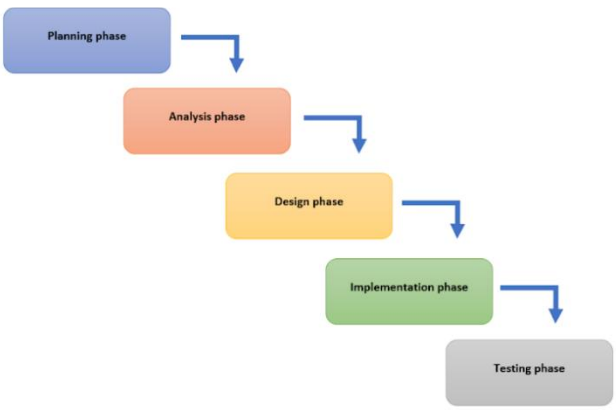

Figure 1: Waterfall Development Methodology

According to figure 1, Waterfall Development in this project includes 5 phases which are planning phase, requirement analysis phase, design phase, implementation phase, and testing phase.

A. *Planning phase*

In this phase, the project's objectives, scope, and deliverables will be defined, as well as the main stakeholders and their requirements. A project schedule will be created, outlined the necessary timelines, and resources for completion.

B. Requirement Analysis Phase

In this phase, the collected information and requirements will be analysed to determine whether these requirements are functional or non-functional. The functional requirement is the process that the system must do. In contrast, the non-functional requirement is the process of how well the system performs, such as usability, security, and performance. After analysing the requirements, it will be documented in the list of requirements. Then, the use case diagrams, activity diagrams and class diagrams can be designed to understand the system's flow.

C. Design Phase

A structural diagram of the monitoring tool's system will be presented in this phase. Low-fidelity prototypes and high-fidelity prototypes will be used to design the monitoring tools. The low-fidelity prototype will structure the core functionality of the monitoring administration, the information architecture, and the flow between pages. The high-fidelity prototype will present the final look of the monitoring administration.

D. Implementation Phase

In this phase, the monitoring tool will be developed and perform the expected functionality to ensure it can function as planned. This monitoring will include GUI (Graphical User Interface) in Java programming language and Stanford Core Natural Language Processing will be used to keep all the project's data.

E. Testing Phase

This phase will be the project evaluation part when the project has been completed. The completed monitoring tool will be tested to ensure it can work properly. "UUM Student Satisfaction Analysis Monitoring Tool" will be evaluated by the management authority of each department in UUM to ensure the objective and scope are fulfilled. In order to evaluate the monitoring tool, USE questionnaire was chosen to conduct the user evaluation to look forward to the user's experience, satisfaction, understanding, and effectiveness when using a model.

2. The Proposed "UUM Student Satisfaction Analysis Monitoring Tool"

This section will represent the proposed algorithm model of "UUM Student Satisfaction Analysis Monitoring Tool". The information pertaining to this section will be displayed in Figure 2.

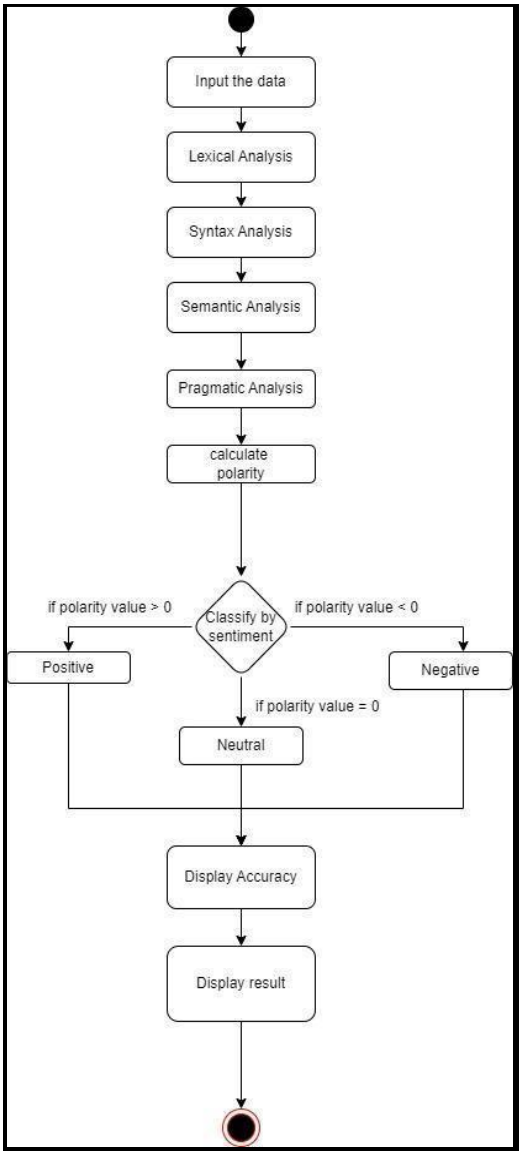

Figure 2: Flow Chart of "UUM Student Satisfaction Analysis Monitoring Tool"

The procedure of "UUM Student Satisfaction Analysis Monitoring Tool" is shown in Figure 2 as it begins with the user input the data from social media text. Inputted text undergoes lexical analysis, which involves breaking it down into individual words or tokens. Subsequently, syntax analysis is employed to comprehend the organizational structure of the input. Following this, semantic analysis comes into play, encompassing the interpretation of context, identification of word connections, and extraction of intended meanings. Pragmatic analysis is utilized to determine the communication's objective and the resulting implications. In the subsequent phases, sentiments are categorized into specific classes, polarity scores are calculated through sentiment analysis algorithms, and the percentage accuracy of positive, negative, and neutral phrases is displayed. Ultimately, the conclusive result is presented as the sentiment score and sentiment class. Figure 3 and Figure 4 displayed the final algorithm model of "UUM Student Satisfaction Analysis Monitoring Tool" based on the procedures above.

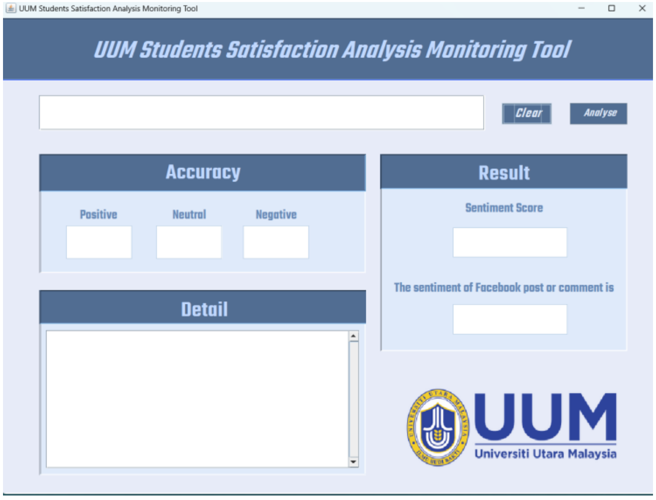

Figure 3: "UUM Student Satisfaction Analysis Monitoring Tool" before analysis the post and comment

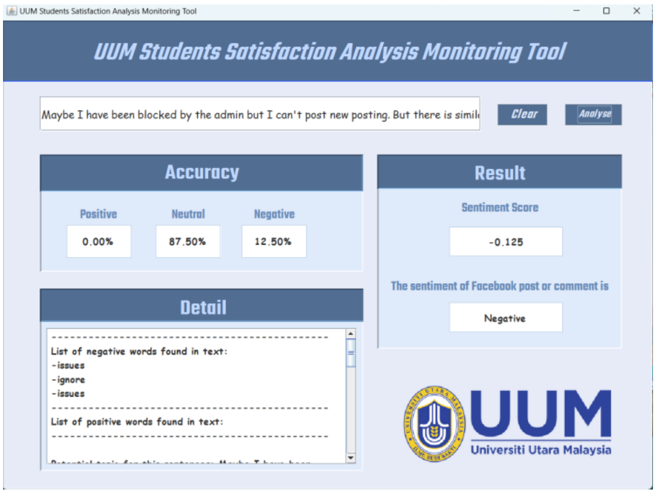

Figure 4: "UUM Student Satisfaction Analysis Monitoring Tool" after analysis the post and comment

IV. Results and Discussion

This section describes the method of evaluation used to effectiveness of the proposed sentiment analysis monitoring tool of this study. In order to accurately evaluate the effectiveness of the proposed model and ensure its alignment with the project's objectives and scope, the USE Questionnaire is employed. The USE questionnaire was chosen because it helps the developer know the user's experience, satisfaction, understanding, and effectiveness when using a model. This questionnaire consists of four dimensions: ease of use, usefulness, satisfaction, and ease of learning. Each of the dimensions had some simple, direct, trusted, and clear statements to help the respondent give an opinion easily. It also helps the moderator quickly get the result, so the moderator can clearly determine the experience and feedback from the respondent.

The UUM Student Sentiment Analysis Monitoring Tool" is developed to help management authorities and staff at UUM understand the analysis result and the details necessary to implement the necessary solution to solve real-life problems and improve the management of the university. To achieve the purpose and propose the outcome of the monitoring accurately, the evaluation was conducted with 20 management officials and employees from the Chancellery Department, Perpustakaan Sultanah Bahiyah, and School of Computing at the University Utara Malaysia (UUM) by stratified sampling technique. Respondents or test participants will be met physically by the moderator and have some field testing and evaluation interaction in a one-on-one session to ensure the participants are accurately selected.

The procedure of evaluation began with the moderator having a field testing briefing with the test respondent to explain the steps using the UUM Student Sentiment Analysis Monitoring Tool to make sure that participants understood the function of the monitoring tool. After briefing with respondents, they will test the monitoring tool based on the explained steps, and they will be given a questionnaire to answer the agreement statement based on four dimensions: ease of use, usefulness of UUMSSAMT, ease of learning, and satisfaction of UUMSSAMT.

1. *Evaluation Strategy*

A. Usability Evaluation

The first dimension is the evaluation of ease of use. In this part, the respondent will give a response about their experience when using the UUM Student Satisfaction Analysis Monitoring Tool, and these responses will reflect the usability of the UUM Student Satisfaction Analysis Monitoring Tool. Subsequently, the usefulness of the UUM Student Satisfaction Analysis Monitoring Tool is explained in terms of how beneficial and effective the tool is in fulfilling its purpose. Moreover, the next section is on ease of learning. The respondents will describe how quickly they were able to understand and navigate the monitoring tool. The fifth section is satisfaction with UUMSSAMT, and it is to collect user responses on user satisfaction while using the monitoring tool created. The details of the mapped questionnaire are presented in Table 1 until Table 4.

TABLE I. EASE TO USE

No	Statement	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
		1	2	3	4	5
1	It is easy to use.					
2	It is simple to use.					
3	It is user friendly					
4	It is flexible					
5	Using it is effortless.					
6	I can use it without written instructions.					
7	I don't notice any inconsistencies as I use it.					

TABLE II. USEFULNESS OF UUMSSAMT

No	Statement	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
		1	2	3	4	5
1	It helps me be more effective.					
2	It helps me be more productive					
3	It is useful.					
4	It gives me more control over the activities in my life.					
5	It makes the things I want to accomplish easier to get done.					
6	It meets my needs.					
7	It saves me time when I use it.					
8	It does everything I would expect it to do.					

TABLE III. EASE TO LEARNING

No	Statement	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
		1	2	3	4	5
1	I learned to use it quickly					
2	I easily remember how to use it.					
3	It is easy to learn to use it.					
4	I quickly became skillful with it.					

TABLE IV. SATISFACTION OF UUMSSAMT

No	Statement	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
		1	2	3	4	5
1	I am satisfied with it.					
2	I would recommend it to a friend.					
3	I feel I need to have it.					
4	It works the way I want it to work.					
5	It is wonderful.					
6	It is fun to use.					
7	It is pleasant to use.					

This questionnaire will use a 5-point Likert scale to examine the respondent feedback on the evaluation of the "UUM Student Sentiment Analysis Monitoring Tool" in different dimensions. The respondent will examine from 1 to 5 (strongly disagree to strongly agree) the monitoring tool as shown in Table 5 below.

TABLE V. SCALE OF LIKERT (5 POINT)

Scale	**Description**
1	Strongly Disagree
2	Disagree
3	Neutral
4	Agree
5	Strongly Agree

The questionnaire also provides a section on the list of aspects, to which respondents are free to give any positive or negative aspect of the UUM Student Satisfaction Analysis Monitoring Tool with their experience, and it is not compulsory. The list of aspects will help identify the strengths and weaknesses of the monitoring tool and help the developer improve it in the future.

B. Unit Testing

Unit testing in this section will describe the accuracy of sentiment analysis. Unit testing is the process of evaluating and validating the validity of elements of the UUM Student Satisfaction Analysis Monitoring Tool system. It requires the creation of test cases to determine whether the sentiment analysis algorithm correctly predicts the sentiment of a given text input. By conducting unit tests, it can ensure that the sentiment analysis system generates accurate results by comparing the expected sentiment output with the actual sentiment output, thereby identifying, and resolving any potential errors or inconsistencies in the analysis process. There are 2 categories of sentiments needed to detect which are positive sentiment and negative sentiment in UUM Student Satisfaction Analysis Monitoring Tool. To identify the performance of the UUM Student Satisfaction Analysis Monitoring Tool, posts and comments from N.E.W.S.E.E.D. on Facebook will be chosen as sample data. N.E.W.S.E.E.D. on Facebook is a private Facebook group that allows UUM students, lecturers, and staff to have some interaction or communication about the university experience without any prescribed topics. The sample size to detect accuracy is 46 because the sample was carefully chosen to reflect a range of viewpoints from various demographics, educational backgrounds, and university life experiences. Besides, the collection and analysis of data from social media platforms might take some time. A smaller sample size makes it easier to thoroughly assess and evaluate each post and comment, ensuring a complete comprehension of the students' opinions. The details of unit testing techniques is shown in Table 6.

TABLE VI. DATASET OF SENTIMENT

Category	Sample size
Positive	3
Negative	43
Total	46

To measure the accuracy of the sentiment analysis algorithm, F1 score was chosen in this project. The F1 score is a measurement task to measure the performance of UUM Student Satisfaction Analysis Monitoring Tool. It combines precision and recall into a single metric that represents the model's accuracy. Precision determines how well the UUM Student Satisfaction Analysis Monitoring Tool avoids false positives. A higher precision indicates fewer false positives. Recall indicates how well the UUM Student Satisfaction Analysis Monitoring Tool identifies true positives and avoids false negatives. A higher recall indicates fewer false negatives. The formula for calculating the F1 score, precision and recall are shown below:

$$Precision = \frac{TruePositives}{(TruePositives + FalsePositives)}$$

$$Recall = \frac{TruePositives}{(TruePositives + FalseNegatives)}$$

$$F1\ score = 2 * \frac{(precision * recall)}{(precision + recall)}$$

2. *Results and Discussions*

A. *Evaluation result*

In this section, the study's findings are thoroughly discussed from a broad perspective. As previously mentioned, the result starts with discussing the evaluation of ease of use when using the UUM Student Satisfaction Analysis Monitoring Tool. The second result will present the usefulness of the UUM Student Satisfaction Analysis Monitoring Tool. Moreover, the next result highlights the ease of learning the monitoring tool. The last result delivered satisfaction with UUMSSAMT. In addition, the accuracy of UUM Student Satisfaction Analysis Monitoring Tool to analyse the sentiment with F1-score was also discussed. The details of the discussion will be addressed in the subsection below.

Ease to use

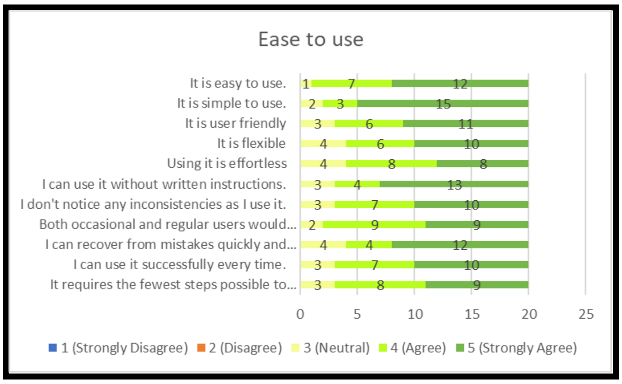

Figure 5: Bar Chart of ease to use

Based on the data presented in Figure 5, a substantial majority of respondents express a strong agreement regarding the user-friendliness of the UUM Student Satisfaction Analysis Monitoring Tool. Specifically, 12 respondents (60%) strongly agree, while seven respondents (35%) agree with this assertion. A mere one respondent (5%) maintains a neutral stance, and no instances of strong disagreements or disagreements are reported. This favourable inclination can be attributed to the provision of explicit instructions and supportive assistance by the moderator, which collectively contribute to fostering a positive and intuitive user experience.

A noteworthy observation indicates that the UUM Student Satisfaction Analysis Monitoring Tool is widely perceived as uncomplicated by the majority of participants. Out of the surveyed individuals, 15 respondents (75%) strongly agree with this notion, while three respondents (15%) simply agree. Merely two respondents (10%) retain a neutral standpoint, and no instances of strong disagreements or disagreements arise. This positive trend strongly implies that the tool possesses an innate intuitiveness and effectively accommodates user needs.

In a similar vein, a significant majority of respondents assert a strong agreement regarding the high level of user-friendliness associated with the UUM Student Satisfaction Analysis Monitoring Tool. Specifically, 11 respondents (55%) strongly agree, while six respondents (30%) express agreement with this statement. However, three respondents (15%) express disagreement, but no instances of strong disagreements or disagreements are reported. Respondents acknowledge and value the tool's simplicity and absence of technical complications.

The majority of participants strongly agree that the UUM Student Satisfaction Analysis Monitoring Tool exhibits a notable level of flexibility. Among the surveyed individuals, 10 respondents (50%) strongly agree, while six respondents (30%) agree with this sentiment. Four respondents (20%) maintain a neutral stance, and no instances of strong disagreements or disagreements are observed. This positive trend reflects the satisfaction of respondents with the tool's adaptability and utility.

A significant number of respondents strongly agree that utilizing the UUM Student Satisfaction Analysis Monitoring Tool requires minimal effort. Within the participant group, eight respondents (40%) agree, and an equal number of respondents (40%) express a strong agreement with this statement. Additionally, four respondents (20%) remain neutral, and no instances of strong disagreements or disagreements are documented. The ease of use may be attributed to respondents' prior familiarity and experience with similar tools.

The overwhelming majority of respondents strongly concur that they can proficiently utilize the UUM Student Satisfaction Analysis Monitoring Tool sans the need for written instructions. A total of thirteen respondents (65%) expresses a strong agreement, while four respondents (20%) agree. Three respondents (15%) maintain a neutral standpoint, with no instances of strong disagreements or disagreements being indicated. This ease of use can be attributed to the tool's judicious employment of clear labelling and a logical arrangement.

A substantial proportion of participants share the opinion that they encounter no inconsistencies while operating the UUM Student Satisfaction Analysis Monitoring Tool. Ten respondents (50%) agree, whereas seven respondents (35%) strongly agree with this assertion. Three respondents (15%) remain neutral, and no instances of strong disagreements or disagreements are reported. This perception of stability and reliability further solidifies the tool's standing.

An overwhelming majority of respondents exhibit a strong agreement that both occasional and regular users would find the UUM Student Satisfaction Analysis Monitoring Tool highly appealing. Nine respondents (45%) strongly agree, while an equivalent number of respondents (45%) express agreement with this statement. Two respondents (10%) remain neutral, with no instances of strong disagreements or disagreements being reported. The respondents greatly value the tool's advantageous features and its direct relevance to their specific requirements.

The majority of participants strongly agree that they possess the ability to swiftly and effortlessly recover from mistakes while utilizing the UUM Student Satisfaction Analysis Monitoring Tool. A total of twelve respondents (60%) strongly agrees, accompanied by four respondents (20%) expressing agreement. Four respondents (20%) adopt a neutral stance, and no instances of strong disagreements or disagreements are expressed. The respondents exhibit confidence in their capacity to promptly rectify any errors.

According to the data delineated in Figure 5, the majority of respondents hold the view that they consistently achieve successful outcomes when employing the UUM Student Satisfaction Analysis Monitoring Tool. Ten respondents (50%) agree, while seven respondents (35%) strongly agree. Three respondents (15%) remain neutral, and no instances of strong disagreements or disagreements are mentioned. The tool's commendable reliability and ability to furnish precise results significantly contribute to the users' accomplishments.

Lastly, most respondents concur that the UUM Student Satisfaction Analysis Monitoring Tool necessitates the fewest possible steps to accomplish their tasks. Nine respondents (45%) agree, while eight respondents (40%) strongly agree with this proposition. Three respondents (15%) remain neutral, and no instances of strong disagreements or disagreements are recorded. The tool's optimal efficiency and streamlined workflow undoubtedly augment the overall user experience.

Usefulness of UUMSSAMT

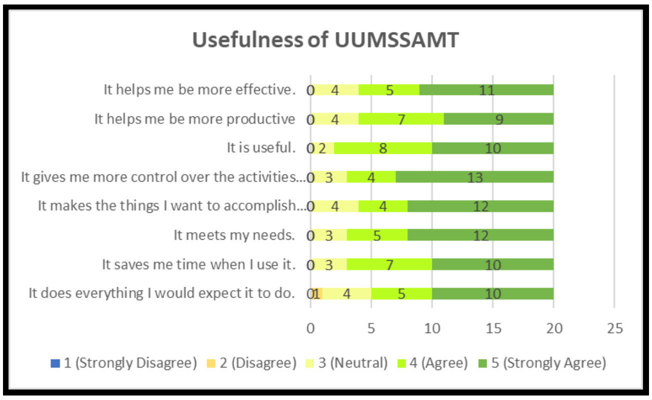

Figure 6: Bar Chart of usefulness of UUMSSAMT

According to the data derived from the survey and presented in Figure 6, the majority of respondents strongly support the notion that the UUM Student Satisfaction Analysis Monitoring Tool enhances their effectiveness. Specifically, 11 respondents (55%) express a strong agreement, while five respondents (25%) agree with this statement. A mere four respondents (20%) remain neutral regarding the tool's ease of use. Notably, there are no instances of strong disagreements or disagreements reported. The analysis demonstrates a positive trend, indicating that the tool's straightforward steps and efficient task completion contribute to its user-friendly nature.

Furthermore, a significant number of respondents strongly agree that the UUM Student Satisfaction Analysis Monitoring Tool increases their productivity. Nine respondents (45%) strongly agree, accompanied by seven respondents (35%) who agree with this statement. However, a minority of respondents express disagreement, with three respondents (15%) disagreeing with the notion that the tool is user-friendly. No instances of strong disagreements or disagreements are reported. This can be attributed to the monitoring tool's valuable capabilities, which aid respondents in gaining a better understanding of their performance and making informed decisions.

The majority of respondents strongly agree that the UUM Student Satisfaction Analysis Monitoring Tool is useful. Ten respondents (50%) express a strong agreement, while eight respondents (40%) agree with this statement. Only two respondents (10%) remain neutral, indicating a minimal level of uncertainty. No respondents indicate disagreement or strong disagreement with the statement. The positive trend observed suggests that respondents are more inclined to find the UUM Student Satisfaction Monitoring tool useful when it provides substantial information.

Additionally, a majority of respondents agree that the tool provides them with a greater sense of control over their activities. Thirteen respondents (65%) agree, while four respondents (20%) maintain a neutral position on this statement. Three respondents (15%) strongly agree, signifying their belief in the tool's empowering nature. No instances of strong disagreements or disagreements are reported. The positive trend can be attributed to the topic addressed, which may instil a sense of empowerment and control in the respondents.

Most respondents agree that the UUM Student Satisfaction Analysis Monitoring Tool simplifies the accomplishment of their desired tasks. Twelve respondents (60%) agree with this statement, while four respondents (20%) express agreement. The number of respondents who remain neutral is also four (20%), indicating a level of indecision. No instances of strong disagreements or disagreements are observed. This is mainly due to the tool's ability to save users' time by automating procedures or providing easy access to resources and information.

Moreover, a majority of respondents agree that the UUM Student Satisfaction Analysis Monitoring Tool adequately meets their needs. Twelve out of 20 respondents (60%) express agreement, while five respondents (25%) strongly agree with this statement. Three respondents (15%) maintain a neutral standpoint. No instances of strong disagreements or disagreements are recorded. The tool has demonstrated its effectiveness in fulfilling the respondents' requirements, providing valuable insights, and facilitating easy access to necessary information.

The majority of respondents, consisting of 10 individuals (50%), strongly agree that the UUM Student Satisfaction Analysis Monitoring Tool saves them time during usage. Additionally, seven respondents (35%) express agreement with this statement. The number of respondents who remain neutral is three, accounting for 15% of the total number of respondents. No respondents strongly disagree or disagree with the statement that the tool can be utilized without written instructions. This indicates that users do not require extensive support or instructions to navigate the software and its interface.

Based on the data collected and presented in Figure 6, a majority of respondents agree that the UUM Student Satisfaction Analysis Monitoring Tool fulfils all their expectations. Ten out of 20 respondents (50%) agree with this statement, while five respondents (25%) strongly agree. Four respondents (20%) adopt a neutral stance, and only one respondent (5%) disagrees that the tool meets their expectations. No instances of strong disagreement are reported. The observed trend is positive, suggesting that the tool has met the respondents' needs, provided crucial information, and aided in decision-making.

Ease of Learning

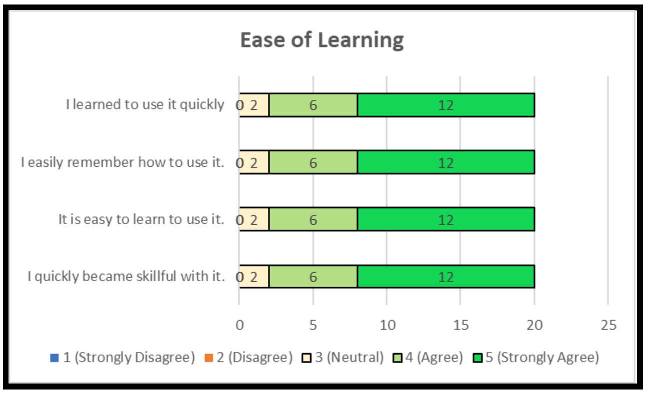

Figure 7: Bar Chart of ease of learning

Based on the data presented in Figure 7, a significant majority of respondents agree with the statement that they were able to learn how to use the UUM Student Satisfaction Analysis Monitoring Tool quickly. Specifically, 12 out of 20 respondents (60%) strongly agree with this statement, while six respondents (30%) agree with it. Only two respondents (10%) maintain a neutral position regarding the speed of learning the tool. Notably, there are no instances of strong disagreements or disagreements reported. The observed trend can be attributed to the provision of clear guidance and effective instructional materials, which assisted respondents in comprehending the tool's capabilities and acquiring proficiency in its usage.

According to the statistics, the majority of respondents strongly agree that they have no difficulty in remembering how to use the UUM Student Satisfaction Analysis Monitoring Tool. A total of 12 respondents (60%) strongly agrees with this statement, while six respondents (30%) agree with it. Two respondents (10%) remain neutral regarding the ease of remembering how to use the tool. No respondents strongly disagree or disagree with this statement. The intuitive interface of the tool likely enhances the user experience and aids users in retaining the knowledge of how to effectively use the monitoring tool.

A majority of respondents express a strong agreement with the statement that the UUM Student Satisfaction Analysis Monitoring Tool is easy to learn. Specifically, 12 respondents (60%) strongly agree with this statement, while six respondents (30%) agree with it. The number of respondents who remain neutral is two, accounting for 10% of the total number of respondents. No instances of strong disagreements or disagreements are reported. The positive trend depicted in the figure can be attributed to the provision of step-by-step guidance and user manuals that explain the tool's functionalities, thereby facilitating an easy learning process.

Furthermore, 12 respondents (60%) strongly agree that they quickly acquired proficiency with the UUM Student Satisfaction Analysis Monitoring Tool, while six respondents (30%) agree with this statement. The number of neutral respondents is two, representing 10% of the total number of respondents. No respondents strongly disagree or disagree with the statement that they rapidly gained skill in using the tool. This indicates that respondents perceive the tool as user-friendly and intuitive, allowing them to quickly grasp its functions and become proficient users.

Satisfaction of UUMSSAMT

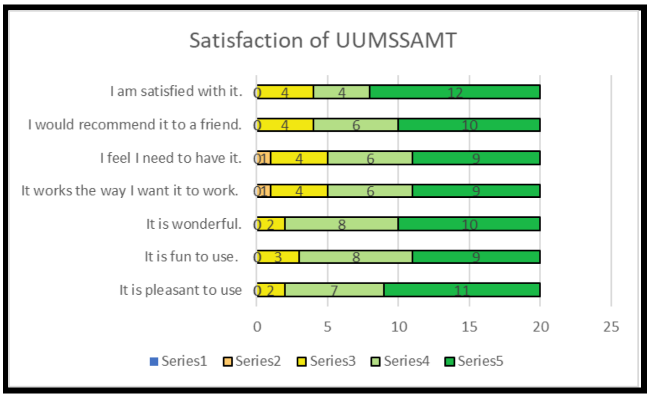

Figure 8: Bar Chart of Satisfaction of UUMSSAMT

Based on the data derived from Figure 8, a significant majority of respondents strongly agree with the statement that they are highly satisfied with the UUM Student Satisfaction Analysis Monitoring Tool. Out of the total respondents, 12 out of 20 (60%) strongly agree with this statement, while four respondents (20%) simply agree with it. Notably, the number of respondents who remain neutral is equal to the number of respondents who agree, accounting for 20% or four individuals. No respondents expressed strong disagreement or disagreement with the notion of being satisfied with the tool. The positive trend observed can be attributed to the tool effectively meeting the expectations and needs of the majority of users.

According to the presented figure, a majority of respondents strongly agree that they would recommend the UUM Student Satisfaction Analysis Monitoring Tool to their friends. Specifically, 10 respondents (50%) strongly agree with this statement, while six respondents (30%) simply agree. Additionally, four respondents (20%) maintain a neutral stance regarding the recommendation of the tool to their friends. No respondents strongly disagree or disagree with this statement. This indicates that the tool receives widespread favourable ratings and fulfils users' expectations, leading them to have a positive inclination to recommend it to others.

A majority of respondents agree with the statement that they perceive a need to have the UUM Student Satisfaction Analysis Monitoring Tool. Nine respondents (45%) agree with this statement, while six respondents (30%) strongly agree. Furthermore, four respondents (20%) remain neutral on this statement. However, there is one respondent who disagrees with the notion of feeling the need for the tool. No respondents strongly disagree with this statement. Overall, the observed trend suggests that respondents generally accept and hold a favourable opinion of the UUM Student Satisfaction Analysis Monitoring Tool.

The UUM Student Satisfaction Analysis Monitoring Tool effectively meets the desired functionalities of a majority of respondents, with nine respondents (45%) strongly agreeing and six respondents (30%) agreeing with this statement. Additionally, only one respondent disagrees, while four respondents remain neutral, indicating a significant level of satisfaction with the tool's usability and effectiveness.

A consensus among respondents reveals that the UUM Student Satisfaction Analysis Monitoring Tool is deemed excellent, as 10 respondents (50%) agree and eight respondents (40%) strongly agree with this statement. Notably, there are no respondents who strongly disagree or disagree, and only two respondents maintain a neutral stance. This overwhelming agreement underscores the tool's ability to meet users' expectations and demands.

The data collected from Figure 8 supports the strong agreement among respondents regarding the enjoyable nature of the UUM Student Satisfaction Analysis Monitoring Tool. Nine respondents (45%) strongly agree, while eight respondents (40%) simply agree that the tool is enjoyable to use. Significantly, no respondents express strong disagreement or disagreement, with only three individuals remaining neutral. This finding can be attributed to the tool's user-friendly design, which encourages exploration and engagement, thereby enhancing the overall user experience.

A substantial number of respondents agree that the UUM Student Satisfaction Analysis Monitoring Tool provides a pleasant user experience. Specifically, 11 respondents (55%) agree, and seven respondents (35%) strongly agree with this statement. While a small minority of two respondents (10%) remain neutral, no respondents strongly disagree or disagree with this notion. The prevalence of positive feedback and recommendations from previous users contributes to this trend, influencing respondents' perceptions and increasing their inclination to agree with the statement.

B. User Testing Result

In this subsection will describe the result of accuracy to measure the performance of UUM Student Satisfaction Analysis Monitoring Tool. Precision and recall are required for the F1-Score calculation of accuracy. Precision and recall are required true positive value, false positive value, and false negative value in calculation. The table below illustrates the true positive, false positive, and false negative values for each sentiment category.

TABLE VII. TABLE OF RESULT TRUE POSITIVE, FALSE POSITIVE, FALSE NEGATIVE, PRECISION, RECALL OF EACH SENTIMENT CATEGORY.

Category	True positive	False positive	False negative	Precision	Recall
Positive	100%	0%	0%	100%	100%
Negative	88.37%	0%	11.63%	100%	88.37%

Based on the data presented in Table 7, true positive of positive sentiment was showing 100% (3) while False positive and false negative are showing 0%. This result indicates that no negative sentiment samples were incorrectly classified as positive, and no positive sentiment samples were incorrectly classified as negative. True positive of negative sentiment was showing 88.37% or 38 of total negative sentiment samples. False positive of negative sentiment is none while false negative sentiment was displaying 11.63% (5). It means some negative sentiment samples were wrongly categorized. Precision of both sentiments are 100% means all the samples identified as positive or negative sentiment were indeed accurate. Recall of positive sentiment is 100% while recall of negative sentiment is 88.37%. It means all instances of positive sentiment were accurately recalled by the algorithm. In contrast, the recall rate for negative sentiment was 88.37%, demonstrating that the algorithm captured the majority of negative sentiment instances but not all of them.

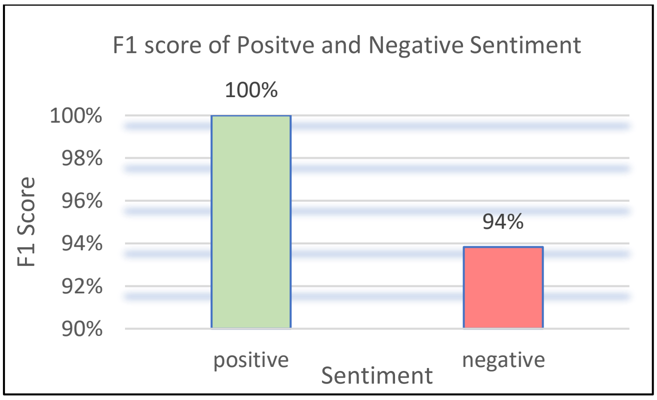

Figure 9: F1 score of positive and negative sentiment detection.

Figure 9 presents the F1 scores for positive and negative sentiment, as evaluated by the proposed algorithm. The F1 score for positive sentiment demonstrates a flawless 100% accuracy in detection, while the F1 score for negative sentiment achieves a commendable 94% accuracy rate. Accurate detection of negative sentiment is challenging due to its various expressions, including irony. The algorithm's reliance on direct word detection without considering grammar structure and context results in decreased accuracy. Understanding linguistic cues and incorporating advanced techniques is crucial for accurately identifying negative sentiment and improving sentiment analysis effectiveness. These findings firmly establish the effectiveness and precision of the proposed sentiment analysis algorithm in distinguishing between positive and negative sentiment.

V. Conclusion

In conclusion, The UUM Student Satisfaction Analysis Monitoring Tool received highly positive feedback from respondents regarding its being user-friendly, providing clear directions, meeting users' needs, saving time, being easy to acquire proficiency, and being enjoyable to use. It was proven that the UUM Student Satisfaction Analysis Monitoring Tool is valuable for analyzing the student's opinion on social media. The outcome enables the management authority of each department and staff in UUM to understand the analysis result and the details to implement the necessary solution to solve the real-life problems and improve the management of the university. While there were some improvements needed further, such as improving the sentiment analysis dictionary, incorporating clear topic detection, and enabling automatic analysis by file, the monitoring tool could become more useful and effective.

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